SERVICE-ORIENTED ARCHITECTURE (SOA) ROADMAP SUPPLEMENTAL GUIDE

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The materials presented in this technical guideline provide specific supplemental information into OUM for the purposes of creating a SOA Roadmap. The SOA Roadmap Supplemental Guide contains the following components:

SOA Roadmap Overview

This component provides an overview of the concepts of SOA Roadmap creation.

SOA Roadmap Supplemental Task Guidelines

The Supplemental Task Guidelines should be used in conjunction with the standard OUM tasks in order to accomplish SOA Roadmap work at the detail task level.

PDF View

You may also prefer to access this information through the **SOA Roadmap Supplemental Guide (PDF View)** located in the Supplemental Guides section of the Supplemental Guidance page.

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PURPOSE

The purpose of this document is to provide an overview of a SOA Roadmap engagement, and to provide implementation teams with *supplemental guidance* for the successful planning and creation of SOA Roadmap.

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AUDIENCE

This document is intended for practitioners who will be delivering SOA Roadmap engagements based on the OUM (Oracle Unified Method) approach.

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HOW TO USE THIS DOCUMENT

This document contains supplemental guidance for planning and executing SOA Roadmap engagements and should be used in conjunction with and not instead of the OUM base method guidance. In many cases, the work that needs to be done for the SOA Roadmap creation is fully contained within the OUM base method task and technique files, therefore, no task supplement is needed for that task.

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LIMITATIONS TO THIS DOCUMENT

This document does not describe OUM in detail &nda the reader must review the documentation and training materials referenced below to gain a basic understanding of each.

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RELATED DOCUMENTATION AND TRAINING

The following related documentation & training provide additional detailed guidance related to OUM implementations and are prerequisites to this document:

- **OUM Method Pack – Desktop Installable or Hosted version**
- **OUM Training and Accreditation Program**

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SERVICE-ORIENTED ARCHITECTURE (SOA) ROADMAP OVERVIEW

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WHAT IS SOA ROADMAP?

A Service-Oriented Architecture (SOA) Roadmap provides the guidance to successfully adopt a service-oriented approach to building IT solutions to support the business. A well constructed SOA Roadmap defines projects that incrementally deliver SOA and provide business value at each step while minimizing the risks associated with SOA adoption. To build such a roadmap requires a solid understanding of the current situation with respect to SOA, a clear vision of what the SOA initiative needs to accomplish to be successful, and a structured approach to defining the iterations that comprise the path to successful SOA adoption.

Background

This SOA Roadmap view is intended to show how to conduct an engagement using OUM to create a SOA Roadmap based on the SOA Roadmap described in the IT Strategies from Oracle (ITSO) guidelines and reference architectures library, Creating an SOA Roadmap. Refer to the Supplemental Guidance page for further information on the IT Strategies from Oracle (ITSO) web site.

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WHEN DO YOU USE THE SOA ROADMAP VIEW?

This view is useful as follows:

- First, if you are planning to create a SOA Roadmap based on the ITSO Creating an SOA Roadmap, then this view helps you see which OUM tasks align with the document. Viewing the SOA Roadmap Supplemental Task Guidelines, you can easily associate the guidelines provided in the ITSO Creating an SOA Roadmap back to the OUM task guidelines. Doing so will help you provide a good foundation for any later programs or projects that may be initiated as a result of the roadmap. This is particularly useful if these programs or projects will use OUM, and they can better relate to the material produced from the roadmap, as the outcome itself is aligned with OUM. This allows work product artifacts of the SOA Roadmap effort to be consumed by the actual delivery projects, as necessary.

- Second, this view may be helpful if you plan to execute an OUM project that has been initiated from a SOA Roadmap made using the ITSO Creating an SOA Roadmap. In this situation, use this view during the Start Up phase to see how the material produced from the roadmap can be mapped to OUM tasks.

The main objective of this view is to **minimize the gap between SOA Roadmap engagements and subsequent implementation projects**.

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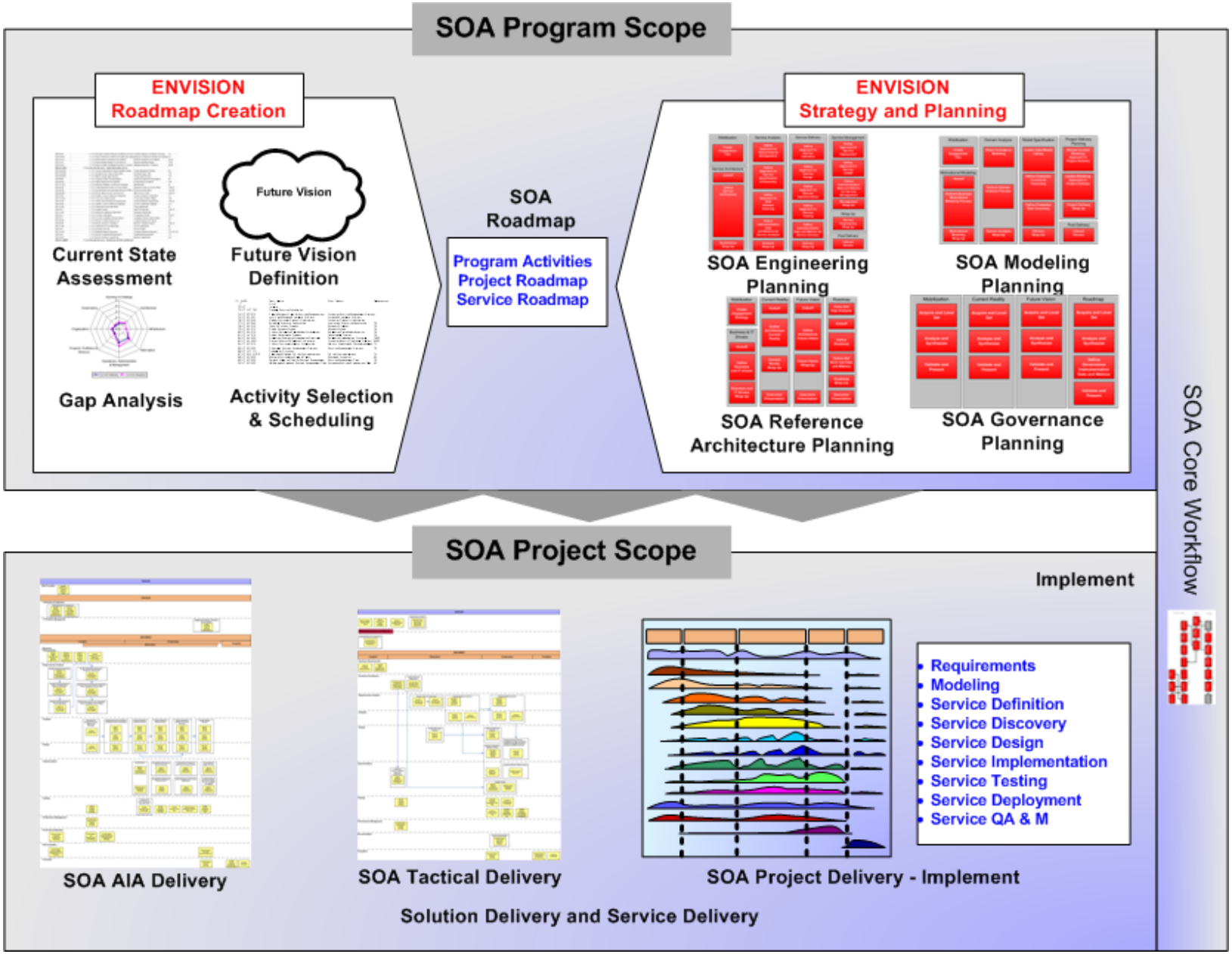
SOA ROADMAP OVERVIEW

A SOA Roadmap provides guidance to the SOA initiative allowing multiple projects to progress in parallel yet remain coordinated. This approach ultimately results in a common end goal that provides value greater than the sum of the individual projects. The SOA Roadmap consists of three fundamental parts:

- program-level efforts
- a portfolio of projects that build specific business solutions
- a portfolio of shared services

The relationship for these three fundamental parts within OUM is illustrated in the figure below:



The program-level efforts create the processes and assets that are leveraged across all of the individual projects. Examples include the SOA Reference Architecture, governance policies and processes, standards, metrics, training and mentoring, service engineering method, etc. The program level efforts provide and enforce the necessary consistency required to succeed at SOA adoption. A delicate balance needs to be struck between too little control and too much control. With too little control, SOA adoption will be haphazard at best. Too much control may stifle project teams resulting in lack of compliance to the program. The portfolio of projects is the route to realizing the SOA Reference Architecture, identifying shared services, and delivering measurable business benefit. The projects that make up the portfolio need to be chosen based on their business benefit, risk, and service reuse opportunity. Initial projects drive the service infrastructure build-out and identify the initial shared services. Follow-on projects leverage the service infrastructure and reuse the shared services. Obviously it is the follow-on projects that demonstrate the full value of SOA.

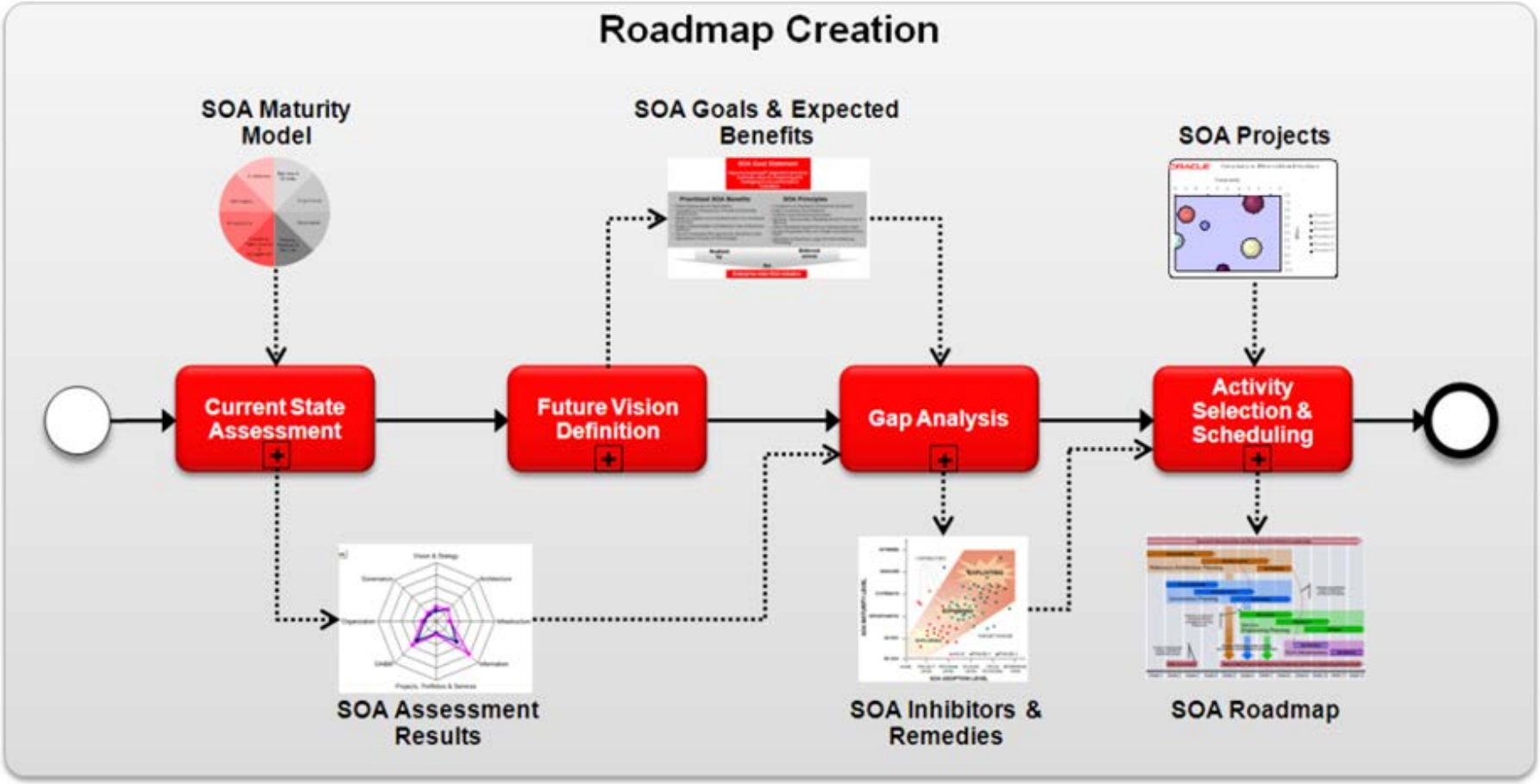
For most companies it is advantageous to separate the service infrastructure deployment and configuration into a separate project from the business solution projects.

This allows the infrastructure project to focus on creating a sound service infrastructure that meets the needs of multiple business solution projects. There will likely be an infrastructure project for each phase of the SOA Roadmap to incrementally deploy the service infrastructure.

The portfolio of shared services creates the catalog of services that is used (and reused) by projects to deliver business solutions. The portfolio of services is separated from the portfolio of projects to eliminate service redundancy and to foster better service reuse by ensuring that each service is designed and constructed based on the needs of more than just a single project. Separating the portfolio of services from the projects also helps prevent tight coupling between the shared service and the business solution that uses the service.

Generally, the most effective planning horizon for a SOA Roadmap is 2-3 years. This could be longer or shorter depending on the planning cycles for each organization. The initial phases (e.g., first six months) of the roadmap contain much greater detail than the later phases. This is appropriate and by design. The SOA journey is a journey of discovery, incremental improvement, and regular course corrections. The SOA Roadmap should be regularly reviewed and updated. The business never stays static, so do not expect the SOA Roadmap to remain static either.

The SOA Roadmap Creation Process



There are four areas in the SOA Roadmap view, each with its own set of tasks.

The areas in the SOA Roadmap follow the standard four steps used within the industry to create roadmaps, i.e., establish the current state, define the future vision, analyze the gap, and define the phases and schedule of the roadmap. It is the particulars within each step of the overall process that provide the uniqueness and value of the approach.

CURRENT STATE ASSESSMENT – The current state is measured based on the Oracle SOA Maturity Model. Using this approach to assess the current state of the SOA initiative provides a consistent measurement scale while keeping the effort focused on capabilities important to SOA success and avoiding the scope creep that frequently undermines current state evaluation efforts. Additional guidance on Current State Assessment can be found in the SOA Roadmap Task Guidelines.

FUTURE VISION DEFINITION – The Future Vision Definition is used to establish the high-level goal and reason for the SOA program. While a fully fleshed out future vision is needed eventually, the initial roadmap creation only requires the high-level vision since the development of the detailed vision can itself be part of the SOA Roadmap. Of course, if the current state of the SOA initiative includes a more detailed future vision, that vision can be leveraged when creating the roadmap. Additional guidance on Future Vision Definition can be found in the SOA Roadmap Task Guidelines.

GAP ANALYSIS – The Gap Analysis evaluates the gap between the current state and the future vision for each of the capabilities. Generally the capabilities exhibiting the largest gap are given highest priority during the roadmap creation. However, part of Gap Analysis also includes evaluating the relative importance for each of the capabilities for the particular organization. Size, organizational structure, existing assets, funding priorities, even politics can significantly impact the relative importance of capabilities. Additional guidance on Gap Analysis can be found in the SOA Roadmap Task Guidelines.

ACTIVITY SELECTION AND SCHEDULING – Finally, we have Activity Selection and Scheduling that uses the output from the Gap Analysis to create a logical ordering for the work to be done. Emphasis is placed on the program-level efforts for the initial phases to establish the assets and processes used across projects and services. Projects are evaluated for business benefit, risk, and SOA applicability and are then prioritized based on that evaluation. The service portfolio is derived from the needs of the prioritized projects. Additional guidance on Activity Selection and Scheduling can be found in the SOA Roadmap Task Guidelines.

The SOA Roadmap creation process is an iterative process. The first iteration may have very limited detail on the project and service portfolios and focus heavily on the program level efforts. As the roadmap is reviewed and updated, additional details are added for the project and service portfolios. As maturity increases and adoption spreads, it is also likely that later phases will include additional program-level activities.

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WHAT OTHER VIEWS RELATE TO THE SOA ROADMAP VIEW?

Refer to the SOA in OUM page for access as well as descriptions for all the SOA views in OUM.

Refer to the Supplemental Guidance page for further information on the IT Strategies from Oracle (ITSO) web site.

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HOW DOES SOA ROADMAP RELATE TO OUM?

The ITSO SOA Roadmap has been mapped to the Envision focus area of Oracle Unified Method to help show which tasks would typically be performed during each of the four areas, from Current State Assessment to Activity Selection and Scheduling.

See the mapping document and SOA Roadmap view for applicable OUM tasks.

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SERVICE-ORIENTED ARCHITECTURE (SOA) ROADMAP SUPPLEMENTAL TASK GUIDELINES

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This document contains OUM supplemental task guidance for the execution of a service based on the SOA Roadmap view.

- The first column of the following table provides a complete list of all tasks in this document for which "supplemental" guidelines are provided as well as a quick link to the "supplemental" task guideline in this document.
- The second column provides a link to the OUM Task Overview.

The task guidelines in this document should be used to supplement the guidelines that appear in the OUM Task Overviews.

SOA Roadmap - OUM Supplemental Task Guidelines (this document) collapse table | expand table	OUM Task Overview
Current State Assessment	
BA.012 Define Business Scope	BA.012 Define Business Scope
BA.015 Conduct Enterprise Stakeholder Assessment	BA.015 Conduct Enterprise Stakeholder Assessment
ER.015.1 Conduct Enterprise Maturity Assessment	ER.015.1 Conduct Enterprise Maturity Assessment
ER.040 Research Enterprise	ER.040 Research Enterprise
ER.080 Obtain Existing Reference Material	ER.080 Obtain Existing Reference Material
ER.015.2 Conduct Enterprise Maturity Assessment	ER.015.2 Conduct Enterprise Maturity Assessment
ER.015.3 Conduct Enterprise Maturity Assessment	ER.015.3 Conduct Enterprise Maturity Assessment
Future Vision Definition	
BA.010 Identify Business Strategy	BA.010 Identify Business Strategy
ER.160.1 Develop Desired Future State	ER.160 Develop Desired Future State
ER.160.2 Develop Desired Future State	ER.160 Develop Desired Future State
ER.110 Perform Benefit Analysis	ER.110 Perform Benefit Analysis
EA.010 Review Architecture Principles, Guidelines and Standards	EA.010 Review Architecture Principles, Guidelines and Standards

Gap Analysis	
ER.165.1 Identify Capability Challenges	ER.165.1 Identify Capability Challenges
ER.165.2 Identify Capability Challenges	ER.165.2 Identify Capability Challenges
ER.165.3 Identify Capability Challenges	ER.165.3 Identify Capability Challenges
ER.167 Determine Remediation Approaches	ER.167 Determine Remediation Approaches
Activity Selection and Scheduling	
ER.130 Produce MoSCoW Improvement List	ER.130 Produce MoSCoW Improvement List
ER.170 Develop Future State Transition Plan	ER.170 Develop Future State Transition Plan
IP.040 Identify Candidate Projects	IP.040 Identify Candidate Projects
IP.050 Prioritize Projects	IP.050 Prioritize Projects
BA.070.1 Identify Candidate Services	BA.070.1 Identify Candidate Services
IP.020.1 Analyze Services	IP.020.1 Analyze Services
BA.070.2 Identify Candidate Services	BA.070.2 Identify Candidate Services
IP.020.2 Analyze Services	IP.020.2 Analyze Services
IP.060 Develop IT Portfolio Plan	IP.060 Develop IT Portfolio Plan

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CURRENT STATE ASSESSMENT

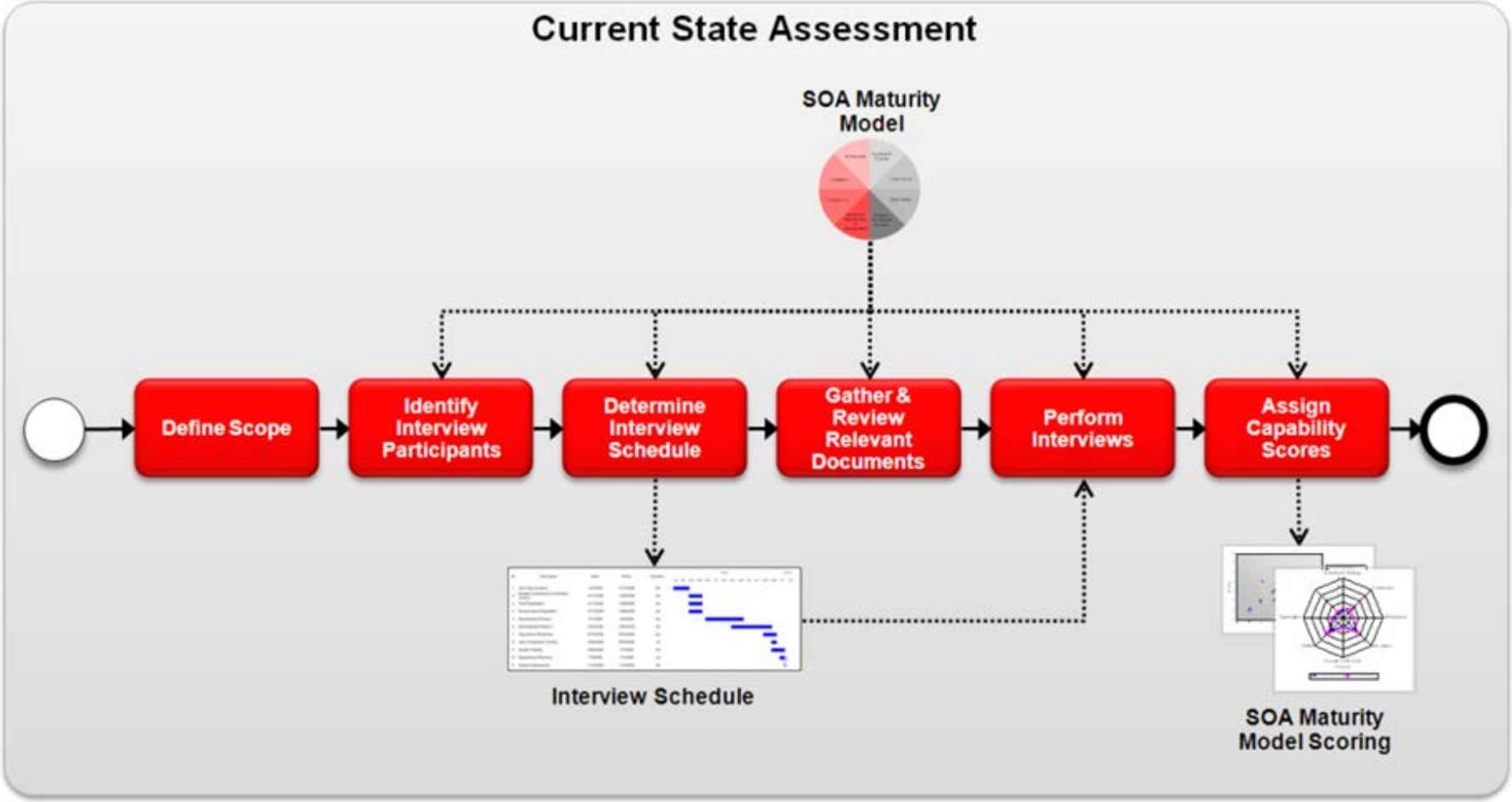
The current state assessment is based on the Oracle SOA Maturity Model. The SOA Maturity Model includes over ninety capabilities that capture the best practices that Oracle has collected over many years working with a wide variety of companies. These capabilities provide the detail necessary to accurately measure and guide the progress of a SOA initiative. Focusing the current state assessment on these specific capabilities ensures a focused scope for the assessment.

For more information on the Oracle SOA Maturity Model, go to the Creating an SOA Roadmap document accessible from the IT Strategies from Oracle (ITSO) web site via the Supplemental Guidance page.

The current state assessment should be tightly time-boxed to ensure timely completion of the assessment. The size and complexity of an organization determines the actual amount of time that must be allocated to the assessment. Nominally, two weeks is the amount of time required. The underlying goal is not to fully capture an IT environment current state; rather it is to evaluate the current state relative to the capabilities that are required to successfully adopt SOA.

At the end of the Current State Assessment you will have completed the SOA Assessment Results. This includes the maturity and adoption score for each of the capabilities in the SOA Maturity Matrix. Additionally, the assessment team will have an understanding of the current state and should have collected known issues and problems that were identified and discussed during the interview process.

An overview of the Current State Assessment workflow is as follows:



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BA.012 Define Business Scope

Define Scope

Before beginning the actual assessment, it is vital that the scope of the assessment is determined and that all involved parties agree to the defined scope. For example, the scope could be limited to a single division or line-of-business within a larger enterprise. Or the scope could be limited to a single geographic location. The scope defines both the scope of the assessment *and*, ultimately, the scope of the roadmap.

When doing this you would typically execute task step 1, Create Business Scope Definition, and depending on the result, execute one or multiple of the remaining steps in the task.

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BA.015 Conduct Enterprise Stakeholder Assessment

Identify Interview Participants

To be able to determine which participants should be interviewed as part of the assessment, perform a stakeholder assessment. With the information you have available, identify the key stakeholders (step 3 of the task), their base concerns, interest and influence (step 5).

Choose the participants that ensure that all capabilities within the SOA Maturity Model can be accurately scored. The following table describes the typical areas of interest and interview participants:

Area of Interest	Typical Participant
Business Objectives	VP of Business Units LOB IT
IT Objectives	CIO VP of Application Development VP of IT Infrastructure
Enterprise Architecture	VP of Enterprise Architecture Enterprise Architect(s)
Program Management	PMO Manager Project Manager(s)
Development Process	Application Architect(s) Business Analyst(s) Development Lead(s) Methodologist Build Manager CM Manager QA Manager
Operations	Director of Operations Administrator(s)
Security	Chieft Security Architect

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ER.015.1 Conduct Enterprise Maturity Assessment

Identify Interview Participants

Based on the stakeholder assessment, determine the final list of participants that should be interviewed as part of the assessment.

Determine Interview Schedule

When you have selected the participants you need to interview, determine an interview schedule. The goal is to limit the length of the Current State Assessment by creating a compacted schedule. It may be necessary to delay the start of the assessment to accommodate the participants' schedules. Ideally, try to fit the all the interviews into approximately a two week period. Time before or between interviews can be productive time spent reviewing documentation (i.e., the next task).

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ER.040 Research Enterprise
ER.080 Obtain Existing Reference Material

Gather and Review Relevant Documents

When you have completed the list of interview participants and know the scope of the assessment, gather as much information as possible about the enterprise situation relevant for the assessment as part of the preparation for the interviews. Execute the tasks steps 3, 4 and 5 of ER.040, but as the purpose is prepare for the interviews, it is not necessary to produce a Discovery Map presentation as a result of this task.

You should also gather any other relevant existing information that may be useful for your interviews preparation. Preferably, the documents to review should be the existing documents that describe various aspects of the current IT environment and SOA initiative. This allows the assessment team to ask more focused questions in the interviews and also provides the opportunity to ask questions about the written material for clarification or to resolve conflicting information. The following list gives some examples of the types of documents that should be gathered and reviewed:

- Balanced Score Card, Strategy Map (or similar business strategy/goals document)
- Enterprise Architecture Document(s)
- Project Management Handbook(s)
- Software Development Process Document(s)
- Operational Process and Procedures Document(s)
- Corporate Security Policies
- Organizational Structure Document
- SOA Program Document(s)

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ER.015.2 Conduct Enterprise Maturity Assessment

Perform Interviews

Before each interview the assessment team should review the SOA Maturity Model to identify capabilities that are particularly relevant for the person being interviewed. It is NOT recommended that the assessment team simply ask a question for each of the capabilities. Rather the interview team should ask open ended questions that allow the interviewee to describe how things are currently done and to identify any problems that currently exists. Remember, the interviewees are the experts on what goes on within the organization being evaluated, so encourage them to explain the current situation.

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ER.015.3 Conduct Enterprise Maturity Assessment

Assign Capability Scores

Once the interviews have been completed and the documents have been reviewed, each of the capabilities in the SOA Maturity Matrix should be scored for both maturity and adoption. These scores provide the raw data that is analyzed during Gap Analysis. The SOA Maturity Model includes a description for each level of maturity and each level of adoption for each capability. When scoring a capability, the scores selected should be the scores where the descriptions of maturity level and adoption level most accurately match the current situation based on the information collected in interviews and from the documents reviewed. Although there is always some level of

subjectivity when measuring capability, the goal is to provide an objective measure. This allows future measurements to be performed by a different assessment team, yet still provide results that can be used to accurately measure progress.

Frequently, when the assessment results are presented there are questions and even disagreements about the score that was assigned. Therefore, it is important that in addition to the score, the assessment team also records the rational for assigning the maturity and adoption scores. This rational could include quotes from interviews or specific sections from the documents that were reviewed.

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FUTURE VISION DEFINITION

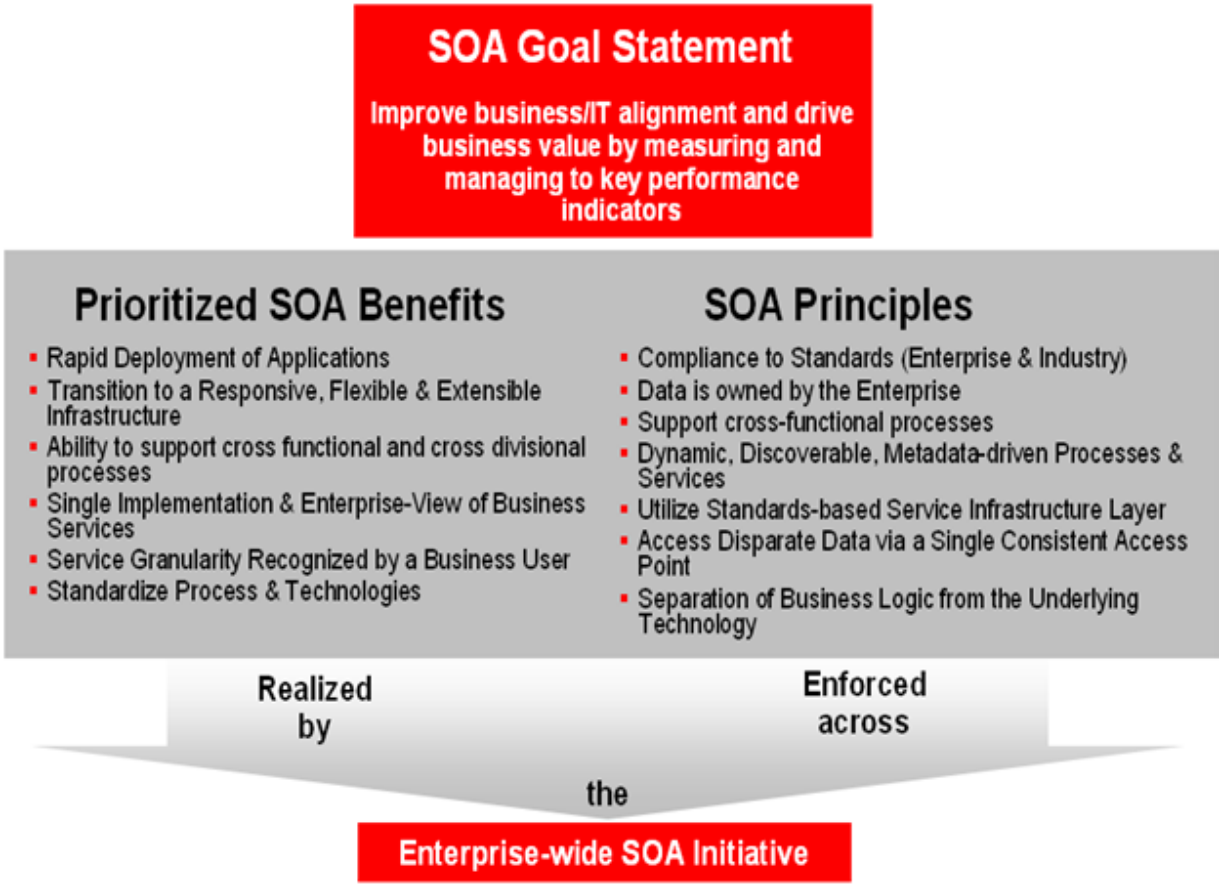
The Future Vision Definition focuses solely on the high-level goals and principles that will be used to guide the entire SOA initiative. Do not attempt to create a detailed future state vision. While a more detailed future vision is required to achieve successful SOA adoption, it is not something that must be created prior to creating the initial SOA Roadmap. The initial phases of the SOA Roadmap may focus on creating the detailed SOA future vision.

The SOA Vision Definition answers the following questions:

1. What is goal of the SOA initiative?
2. What is the organizational scope of the SOA initiative?
3. What are the benefits that SOA is expected to deliver to the organization?
4. What are the guiding principles for the SOA initiative?

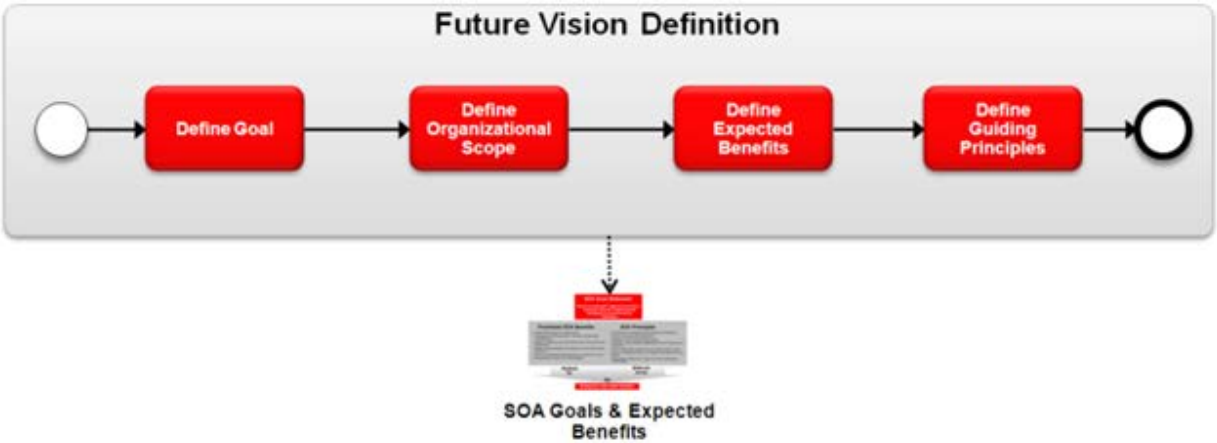
These questions must be answered by the executive(s) leading the SOA initiative. This is accomplished in a facilitated workshop. Nominally, the workshop should take about two hours, but may take longer if there is no pre-existing understanding of SOA or there is substantial disagreement on why SOA is being pursued.

At the end of Future Vision Definition, you have a completed SOA Vision Definition. This contains the scope of the initiative, the expected benefits, and the guiding principles to achieve the goal and the benefits. This vision for the SOA initiative can be captured in a single summary slide and used to educate and align the organization with the SOA initiative. The figure below shows an example of such a summary slide:



This picture clearly shows the goal and scope of the SOA initiative. It also shows that the SOA principles will be enforced across the entire SOA initiative and that the SOA initiative is expected to deliver the prioritized benefits.

An overview of the Future Vision Definition workflow is as follows:



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BA.010 - Identify Business Strategy

Define Goal

Review the Business Strategy and identify areas relevant for SOA. In particular, task steps 4 (Determine Vision, Mission and/or Values) and 6 (Determine Goals, Objectives, Targets and Initiatives) are relevant. Use this as an input to prepare the goals of the SOA initiative. This should help you in answering the question, “What is the goal of the SOA initiative?” above. The goal being defined should be the goal for the SOA initiative after having executed the roadmap.

The recommended roadmap planning horizon is 2-3 years; therefore the goal should be the goal of the SOA initiative 2-3 years from now. The following table provides some example goal statements:

Goal Statement	Score
Develop experience building services to provide a foundation for further investments in SOA.	1
Apply SOA to a limited set of projects to demonstrate the benefits of SOA and build credibility with the business owners.	2
Drive widespread sharing and reuse by consistently applying SOA concepts.	3
Improve business/IT alignment and drive business value by measuring and managing to key performance indicators.	4
Support new business initiatives and exploit new revenue opportunities enabled by SOA.	5

The goal statements in the above table are used to gauge the extent and complexity of the entire SOA initiative and are listed in order of increasing difficulty. It should be obvious that accomplishing the first goal statement is far less difficult than accomplishing the fifth goal statement. Greater organizational maturity is required to achieve the more difficult goals for SOA. Of course, the benefits provided by SOA are commensurably greater as well. The Score column in the table is used (and will be explained) during Gap Analysis.

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ER.160.1 Develop Desired Future State

Define Organizationl Scope

Once you have determined the SOA Goal statement (see BA.010 Identify Business Strategy), you should answer the second question above, “What is the organizational scope of the SOA initiative?” The organizational scope defines which departments, divisions, lines-of-business, etc., are included in the SOA initiative. The most common scopes are either division or enterprise, but other options are possible depending on the company's organizational structure. The following table provides example levels of scope for the SOA initiative.

SOA Initiative Scope	Score
A small number of projects will be applying service-orientation.	1
One or more business units will use SOA to address all of their projects.	2

The SOA initiative will span all business units within a single division.	3
The SOA initiative will span multiple divisions within the enterprise.	4
The SOA initiative will span the entire enterprise.	5

Defining the scope of the SOA initiative is essential to determining a roadmap. With greater scope, the number of organizational boundaries that must be crossed increases. This increases the complexity of the effort, and, therefore, requires greater organizational maturity. The Score column in the table is used (and will be explained) during Gap Analysis.

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ER.160.2 Develop Desired Future State
ER.110 Perform Benefit Analysis

Define Expected Benefit

Finally, you should identify the list of expected SOA benefits that the organization realistically can achieve within the given timeframe, and with a successful adoption of SOA. This will help answer the third question,"What are the benefits that SOA is expected to deliver to the organization?"

There are many different benefits that an organization can realize by successfully adopting SOA. However, not all benefits can be realized in parallel. When creating a SOA Roadmap, emphasis should be placed on the benefits that are highest priority and leave the lower priority benefits for later phases. The list below shows examples of possible benefits from SOA adoption.

- rapid deployment of applications
- unified, simplified data access
- transition to a responsive, flexible, and extensible infrastructure
- flexible architecture that enables business and IT agility
- ability to support cross functional and cross divisional processes
- effective use of external service providers
- single implementation and enterprise-view of business services
- operational improvements
- service granularity recognized by a business user
- skill-set portability
- standardize process and technologies
- standards based integration
- quicker re-composition of business processes
- integration with business partners
- de-commissioning of older applications
- better visibility into business processes
- incremental development and deployment
- business process standardization
- decoupling of applications Productivity gains

These benefits can be used as a starting place to identify the benefits that an organization hopes to achieve via SOA adoption. The above list is by no means a complete list. Part of the SOA Vision Definition is to create a list of possible SOA benefits and prioritize the list. Once a list has been created, the benefits should be prioritized based

on the business and IT objectives of the organization. The easiest way to prioritize the benefits is to assign a high, medium, or low prioritization to each possible benefit. Roughly one third of the possible benefits should be in each prioritization, i.e., it is not helpful to list all the possible benefits as high priority.

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EA.010 Review Architecture Principles, Guidelines and Standards

Define Guiding Principles

The SOA Vision Definition is completed by defining a number of guiding principles. This answers the final question, “What are the guiding principles for the SOA initiative?” The principles should be defined to support the highest prioritized benefits, and should thereby provide enforceable guidance to the SOA initiative. The following list provides some example guiding principles.

- Every service must be documented following company standards.
- Apply a consistent and controlled service identification process.
- Utilize a standards-based service infrastructure.
- Data is owned by the enterprise.
- Enterprise data should be accessed via shared services.
- Avoid use of point-to-point connections.
- Reuse existing services wherever possible.
- Buy before build, reuse before enhance, and enhance before build new.

The guiding principles should be sufficiently clear and detailed so that the principles can be enforced across the entire scope of the SOA initiative and on specific projects that fall under the purview of the initiative. The principles should also serve as a foundation to make more specific decisions in the future.

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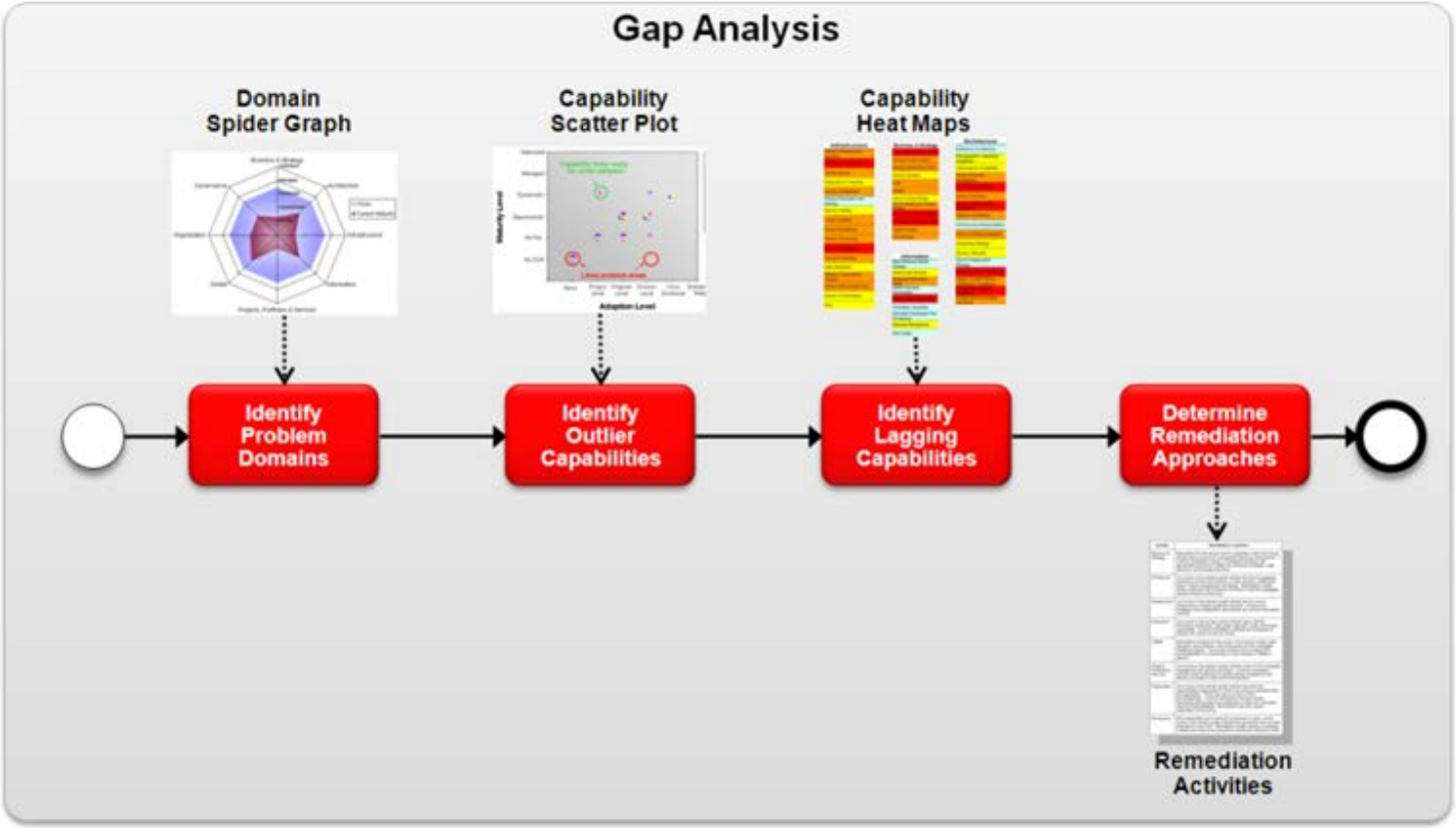
GAP ANALYSIS

During Gap Analysis, the current state of the SOA initiative (as measured during the Current State Assessment above) is compared with the goal for the initiative (defined during Future Vision Definition above). The gap between these two is then analyzed to determine the causes and identify remediation approaches.

The maturity and adoption scores from the Current State Assessment measure the progress of a SOA initiative and, more importantly, identifies specific capabilities that are lacking or lagging and are therefore inhibiting the SOA initiative. The gap between where the organization is currently and where they need to be to achieve their goal is broken down by capability domain (from the SOA Maturity Model) to identify lagging domains. It is further broken down by individual capability to identify specific capabilities that are lacking or lagging. The following diagram illustrates the workflow.



Once the lagging capabilities have been identified, a remediation approach for each of the identified inhibitors is determined from industry best practices and prior Oracle experience.

At the end of the Gap Analysis, you have the Recommendation Activities at domain level. This includes an understanding of which domains and which individual capabilities are inhibiting the successful achievement of the goal of the SOA initiative. Additionally, it includes identified remediation activities that address the lagging domains and capabilities. These remediation activities provide the primary input into the roadmap creation process.

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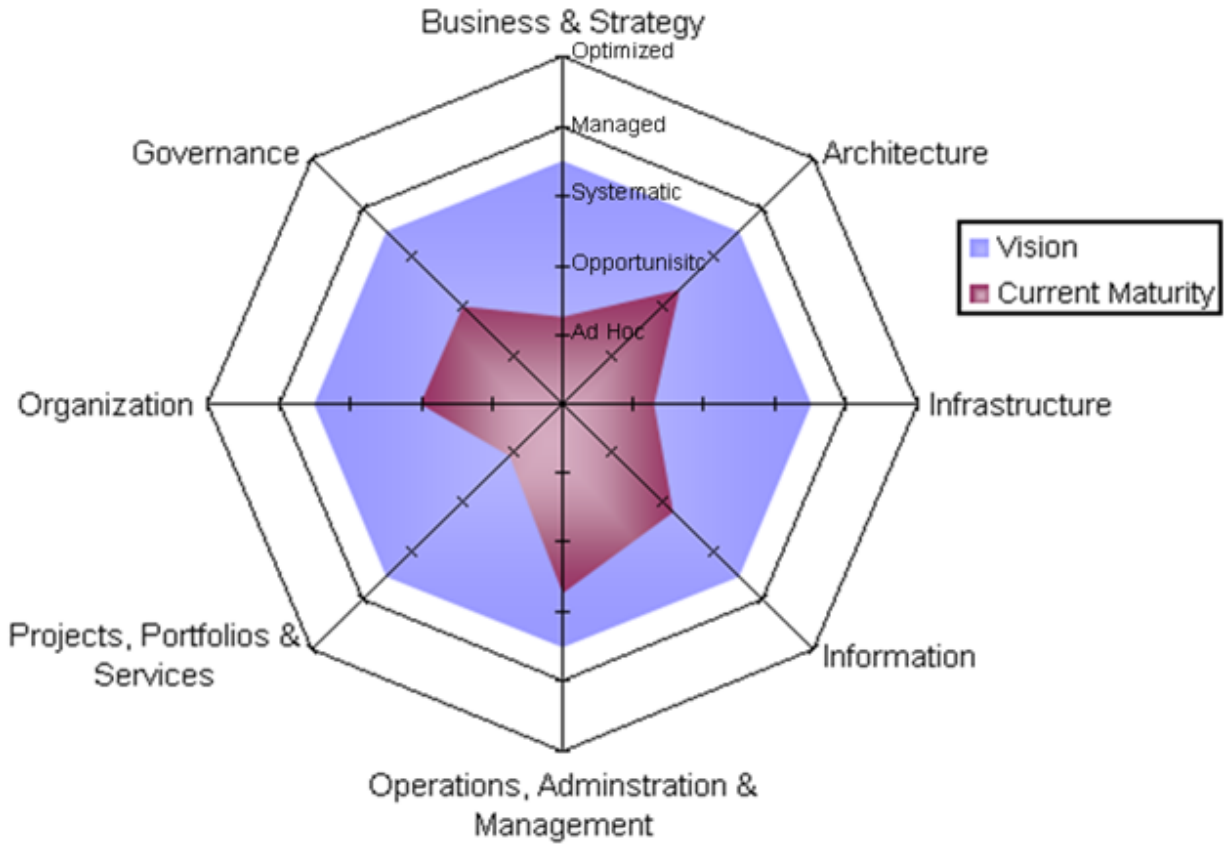
ER.165.1 Identify Capability Challenges

Identify Problem Domains

The first step of Gap Analysis is to identify the domains that exhibit the largest gap between current maturity and the maturity needed to achieve the SOA goal. The gap for the domains can be visually represented by a spider (aka radar) graph as shown below:



Vision versus Current Maturity

The Vision level is determined by the following formula:

$$V_S = \begin{cases} G_S, \text{ if } G_S \geq S_S \\ (G_S + S_S)/2, \text{ if } G_S < S_S \end{cases}$$

Where **Gs** is the score from the goal statement and **Ss** is the score from the scope of the SOA initiative. This formula maps the goal statement to a level of maturity necessary to reach that goal. The scope of the initiative is used as a modifier if the scope score is greater than the goal score, i.e., it takes greater maturity to achieve success on a broader scope. The vision score (**Vs**) is then mapped to the maturity levels in the SOA Maturity Model by applying the simple formula: 1=Ad Hoc, 2=Opportunistic, … , 5=Optimized.

For the example in the spider graph shown above, the Vision level was derived using this formula as follows:

> Goal Statement - Drive widespread sharing and reuse by consistently applying SOA concepts.
>
> Scope Statement - The SOA initiative will span multiple divisions within the enterprise.

So **Gs** = 3, **Ss** = 4, and **Vs** = (3+4)/2 = 3.5. The 3.5 score mapped to the maturity levels yields a vision maturity half way between Systematic and Managed. This Vision level of maturity is what is plotted in the spider graph figure.

The Current Maturity level for each domain is calculated by simply averaging the maturity score for each capability within the domain. This provides an average maturity for each of the eight domains which is then plotted in the spider graph.

Plotting the Current Maturity relative to the Vision (as shown in the spider graph above) provides a visual representation of the gap between where the organization is with respect to SOA and where it needs to be to meet the goal of the SOA initiative. In this example, the graph clearly shows that the Projects, Portfolios & Services domain is the area requiring the most attention followed closely by the Business & Strategy and Infrastructure domains.

This analysis should be fed into Activity Selection and Scheduling. In this example, initial activities in the roadmap should center on addressing the lagging capabilities in the Projects, Portfolios and Services, Business and Strategy, and Infrastructure domains.

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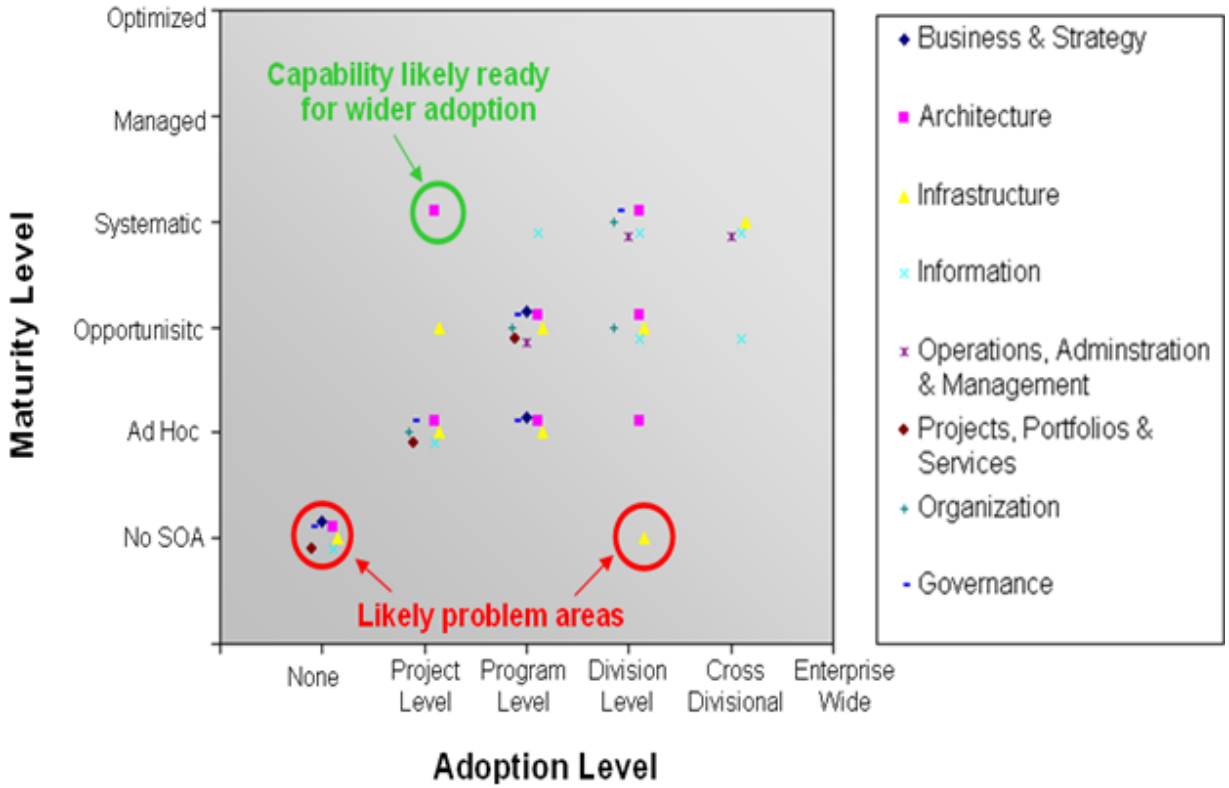
ER.165.2 Identify Capability Challenges

Identify Outlier Capabilities

Outlier capabilities are capabilities where the maturity and the adoption are significantly out of sync. This usually indicates a capability that should receive attention early in the roadmap. The outlier capabilities can be easily detected by plotting the maturity and adoption score for each capability in a scatter plot as shown in the following figure:



Capabilities Scatter Plot

As shown in the above graph, the maturity and adoption scores for capabilities usually fall along the diagonal when plotted against each other. In the above example, there are two scores that are significantly off the diagonal, one well above the diagonal and one well below the diagonal. The capability well above the diagonal is a capability

from the Architecture domain. The capability well below the diagonal is a capability from the Infrastructure domain. The analyst will need to review the actual capability scores to identify exactly which capabilities yielded the outlier points on the graph.

A capability that falls well above the diagonal indicates a capability that is done very well within a relatively small area of the organization. In this example, there is a capability at a Systematic level of maturity done within a single project. Fostering greater adoption of this capability provides an easy win for the SOA initiative i.e. there is no need to develop greater competency for this capability within the organization since it already exists within the organization. Some training or mentoring can spread the ability more broadly within the organization.

A capability well below the diagonal indicates a capability that is done poorly (or in a non-SOA compliant fashion) very broadly. In this example, there is a capability being done at a No SOA level of maturity across the entire division. Corrective action needs to be taken for this capability since if left uncorrected, it will inhibit (and probably already has inhibited) the SOA initiative.

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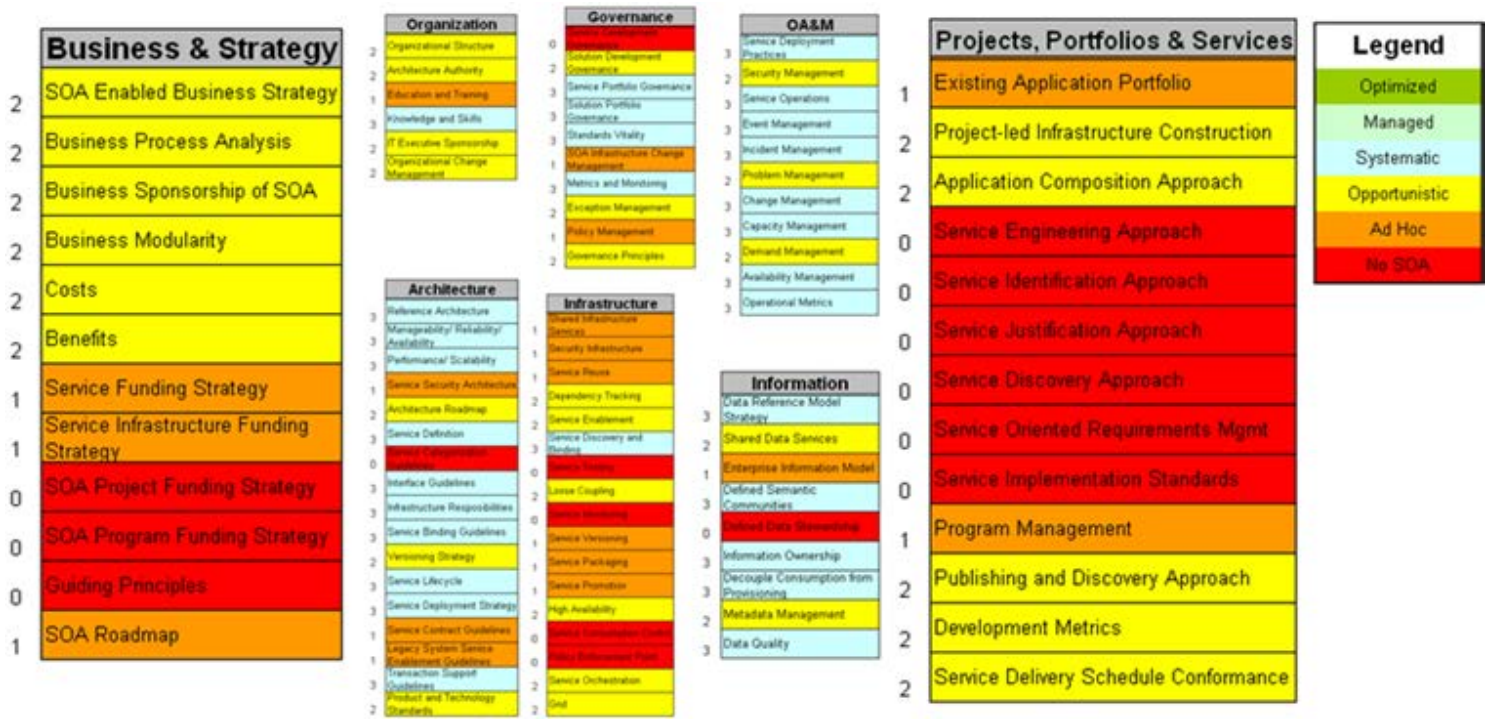
ER.165.3 Identify Capability Challenges

Identify Lagging Capabilities

The lagging capabilities would typically require attention in early phases of the roadmap. These are the capabilities that plot nearer the lower left corner in the scatter plot above. Capability heat maps can be used to visually identify these low maturity capabilities as shown in the following figure:



Capabilities Heat Map

The capabilities heat map colors each of the capabilities based on the maturity score recorded for that capability. The above diagram shows the color coding legend as well as each capability with color coding applied. The capabilities are organized by the domains used in the SOA Maturity Model. The heat map draws immediate attention to the capabilities that require attention. In the above example, there are three capabilities in the Business and Strategy domain (SOA Project Funding Strategy, SOA

Program Funding Strategy, and Guiding Principles) that scored at the No SOA level of maturity. These capabilities should be addressed in early phases of the SOA Roadmap. Likewise, there are six capabilities in the Projects, Portfolios and Services domain that scored at the No SOA level of maturity. These capabilities should also be addressed in early phases of the SOA Roadmap. It should come as no surprise that these two domains were also identified in the spider figure, Vision versus Current Maturity as lagging domains.

You should review each capability with a low score and determine whether it is a top priority, a low priority, or unimportant from a roadmap creation perspective. If a capability is deemed unimportant for a particular organization, it should be removed from the maturity model and the graphics should be regenerated. This should be done with caution. A capability should only be removed if it is clearly not appropriate for the SOA initiative at this organization.

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ER.167 Determine Remediation Approaches

Determine Remediation Approaches

At this point in Gap Analysis, the problem domains have been identified and the problem capabilities within each domain have also been identified. The next step is to identify remedies for each problem domain and capability. Obviously, the remedies depend on the problem being addressed and frequently have some aspect that is organization specific. Therefore, unfortunately, it is not possible to provide a prescriptive approach to determining remediation activities for all 90+ capabilities. However, there are some general guidelines based on the domains that can usually be applied to creating remediation activities, as shown in the table below:

Domain	Remediation Approach
Business & Strategy	Remediation for this domain and the capabilities within this domain usually requires executive management decisions and directives. A common remediation activity is a facilitated workshop with appropriate executives to define the necessary strategies, make decisions, and formulate directives.
Architecture	Low scores in this domain usually indicate the lack of a reference architecture for the SOA initiative, or if the reference architecture exists, it lacks completeness and details. Remediation usually entails workshops with enterprise architects to specify a complete, detailed reference architecture.
Infrastructure	Low scores in this domain usually indicate that the service infrastructure is lacking significant elements. Infrastructure installation and configuration type projects are common remediation activities.
Information	Low scores in this domain usually indicate issues with the information architecture, data quality approach, and/or information stewardship. Common remediation activities are workshops to address the causes for the low scores.
Operations, Administration & Management	Remediation activities for low scores in this domain usually entail definition, documentation, and enforcement of SOA compatible.
OA&M Procedures	Low scores could be due to lacking SOA knowledge/skills or could be due to a low maturity of OA&M in general.
Projects, Portfolios & Services	Low scores in this domain usually indicate a lack of SOA compatible management and delivery processes. Common remediation activities entail workshops to modify existing management and delivery processes to inject SOA best practices.
Organization	Low scores in this domain usually indicate that roles and responsibilities appropriate for SOA have not been instituted within the organization. There may also be a lack of SOA knowledge/skills. Common remediation activities include developing training plans and workshops to define the necessary roles and responsibilities. Remediation may also require organization restructuring.
Governance	Most organizations have existing IT governance in place, so low scores in this domain usually indicate that governance has not been extended to cover SOA. Remediation usually requires a workshop to define and institute the governance extensions required for SOA.

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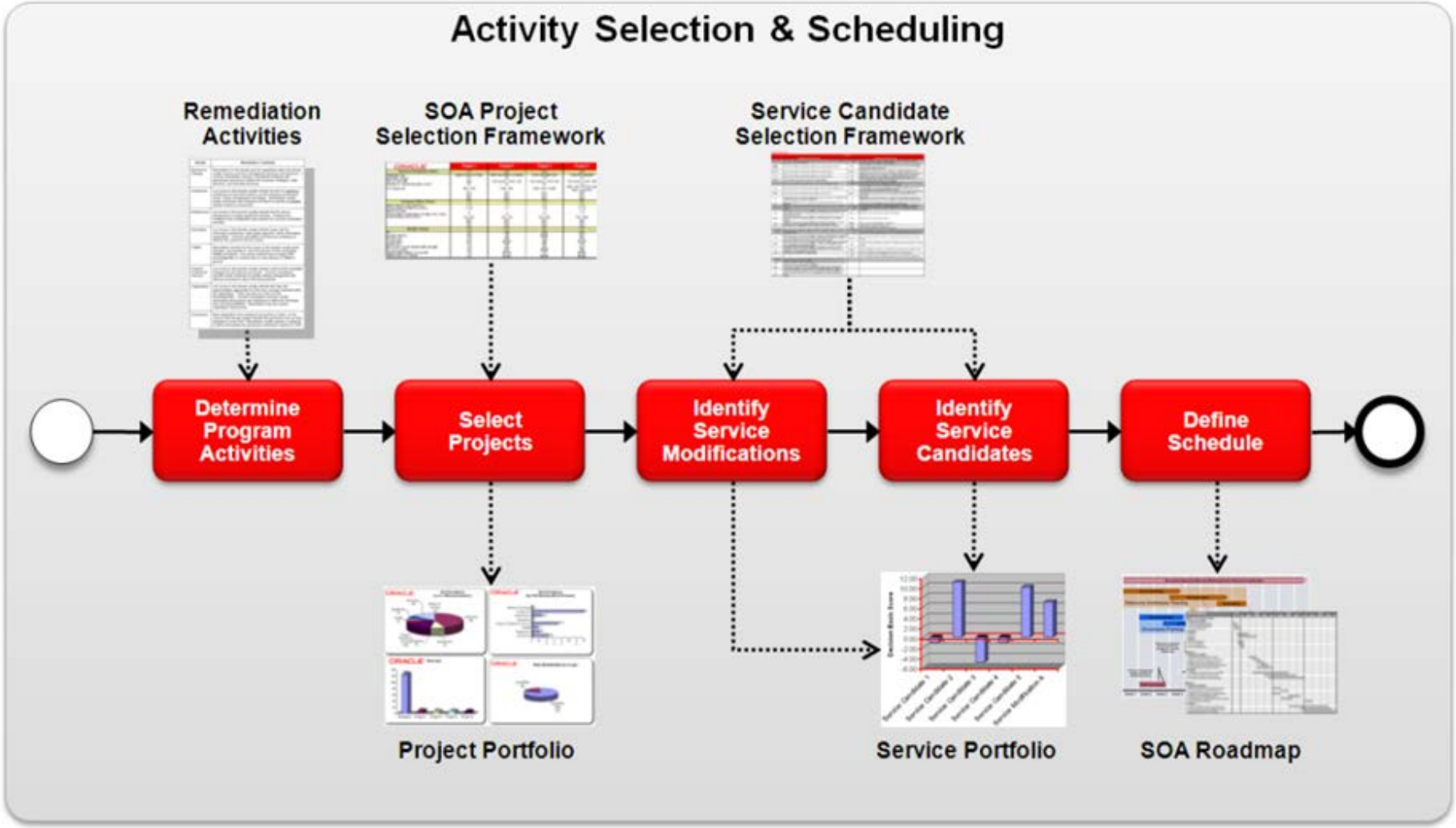
ACTIVITY SELECTION AND SCHEDULING

During Activity Selection and Scheduling, prioritize the remedies that you identified during Gap Analysis and use this to create a plan, called the SOA Roadmap. Projects are evaluated to select the projects that will provide the highest benefit versus the cost, complexity, and risk. The service portfolio is derived from the needs of the projects.

There are five steps in Activity Selection and Scheduling:



At the end of the Activity Selection and Scheduling you have a completed SOA Roadmap. This roadmap includes a detailed initial phase and less detailed subsequent phases. The SOA Roadmap provides guidance for achieving the goal of the SOA initiative via a series of much smaller transitions. Ideally, each of the smaller transitions has its own individual business benefit, but it is frequently necessary to invest a little up front to reap larger benefits later.

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ER.130 Produce MoSCoW Improvement List

Determine Program Activities

The first step when determining the program activities is to prioritize the remediation activities identified in the Gap Analysis. Top priority is usually given to remediation activities that focus on the domain with lowest current maturity score. Top priority remediation activities are usually the first activities in the roadmap since the results from these activities are leveraged across the solution and service delivery efforts.

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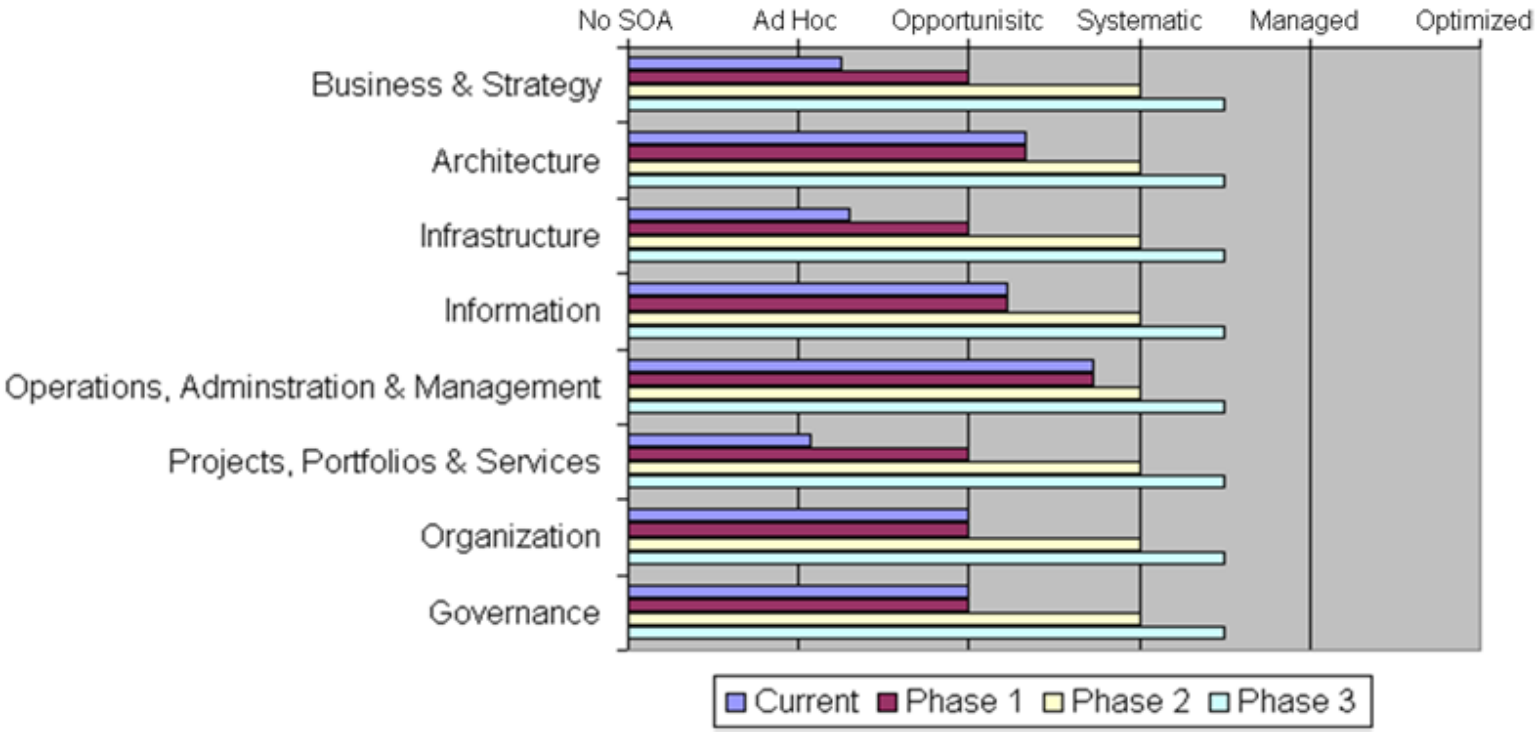
ER.170 Develop Future State Transition Plan

Determine Program Activities

After the program activities have been prioritized, determine when each of the activities should best be executed, and split the SOA Roadmap into phases.

Program-level activities frequently entail changes with wide-ranging impacts. For example, changing the software development process (to inject service-oriented best practices) impacts all development teams within the scope of the SOA initiative.

Organizational changes can be even more taxing. Therefore, it is usually necessary to undertake these changes in a series of iterations. These iterations become the phases of the overall SOA Roadmap. At a high level, this can be shown graphically as illustrated by the figure below:



Increasing Maturity Over Time

The graph illustrates three iterations each increasing maturity of one or more domains until the desired vision level of maturity is achieved. Notice that the first phase

focuses on bringing all domains up to the **Opportunistic** level of maturity. This means that the first phase will include remedy activities for the Business & Strategy, Infrastructure, and Projects, Portfolios & Services domains. Once near parity is achieved across domains, follow-on phases address the eight domains more uniformly to keep the SOA initiative progressing smoothly.

The amount of change introduced by an iteration must not exceed the organization's ability to absorb that change; therefore the scope of each iteration must be carefully planned. Likewise, the duration of each iteration must be long enough to accomplish some meaningful progress, yet remain short enough to minimize risk and maintain a continuous pace of incremental progress.

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IP.040 Identify Candidate Projects
IP.050 Prioritize Projects

Select Projects

Projects should be selected related to the chosen activities where the intention is that each iteration should increase the maturity of the organization while also providing business value. To see what kind of projects already have been identified in the enterprise have a look at the Initial IT Portfolio Plan (IP.010), if available.

Usually the program-level activities do not produce quantifiable business value. Rather, it is the solution development efforts that create measurable business value. Therefore, a successful SOA Roadmap must select projects that:

1. Provide measurable business value.
2. Are amenable to a service-oriented approach.
3. Advance the SOA initiative.

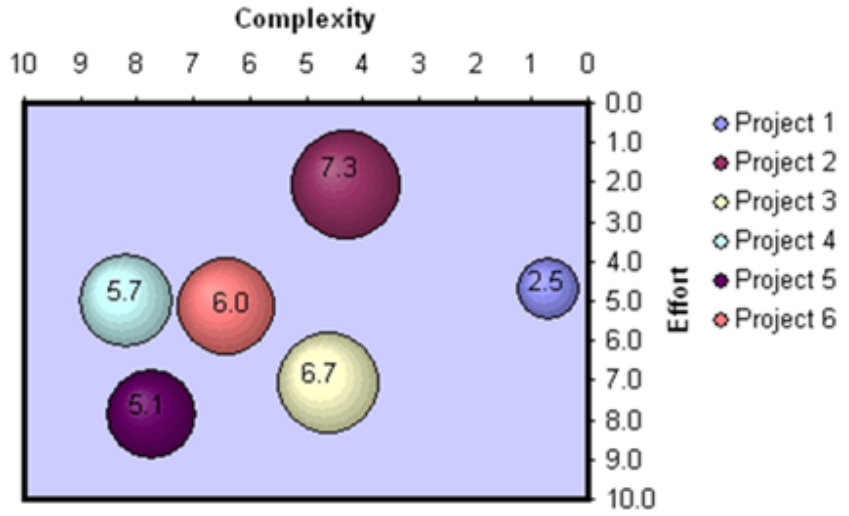
The first of these criteria is a standard requirement for traditional project portfolio planning. The second criteria should be used to eliminate certain projects from further analysis. For example, a project to deploy a maintenance release of a legacy application is not a good candidate for inclusion in the SOA Roadmap. (However, a major upgrade to a legacy application might be a good candidate.)

The third criterion is what is added to traditional project portfolio planning. The additional criteria used to rank projects for inclusion in the SOA Roadmap include:

- benefits in line with the prioritized SOA benefits (defined during Future Vision)
- adherence to the SOA principles (defined during Future Vision)
- functional needs available from already existing shared services
- functional overlap with other projects
- data requirements overlap with other projects
- ability to leverage existing service infrastructure
- infrastructure needs aligned with the SOA Reference Architecture

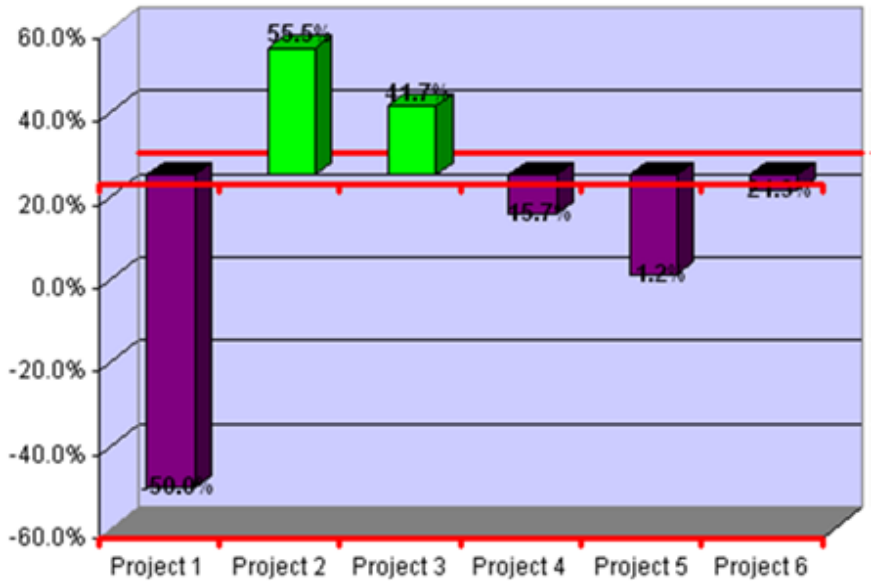
There may also be new risks associated with taking a service-oriented approach to a project. These risks can be specific to the project or might be associated with the SOA initiative itself. For example, if there is no experience or defined best practices for the service infrastructure, a project using the service infrastructure has additional risk associated with that usage. Selecting projects, therefore, requires balancing all these new SOA factors with the traditional factors (e.g. cost, complexity, benefits, risks).

Below, you can see an example graphic that shows the analysis results for six different projects for the three aspects complexity, effort and benefits:



Complexity versus Effort versus Benefit Analysis

The size of the bubble represents the relative benefit that the project would provide and the bubble is plotted versus the complexity and effort for the project. Large bubbles toward the upper right corner are the projects that are expected to provide the greatest benefit for the least effort and complexity. From this analysis, Project 2 is the best choice and Projects 1, 3 and 6 are also contenders for inclusion in the roadmap. The next step is to analyze the risk associated with the projects. This analysis is shown in the next figure:



Benefit to Risk Ratio Analysis

This graph plots the benefit to risk ratio for each of the projects. The higher the number, the greater the benefits from the project as compared to the risks for the project. This graph clearly shows that Project 2 and Project 3 are the best projects to include in the SOA Roadmap. Project 1, which was a contender from the cost and complexity perspective, is clearly not a good choice due to the high risk versus benefit.

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BA.070.1 Identify Candidate Services
IP.020.1 Analyze Services

Identify Service Modifications

Having selected the projects that will become part of the SOA Roadmap, the next step is to identify services and candidate services that should be included in the roadmap. Look at the Service Portfolio (IP.014) to see what kind of services have already been identified, if available.

One of the selection criteria used for the projects was the amount of functionality that was already available from existing services. Now, it is important to see whether it may be necessary to make any modifications to these services to support the new project. There are two main types of changes:

1. Increase the performance or capacity of the service.
2. Implement functional enhancements to the service.

The new project may require that additional capacity or better performance than the existing service currently provides. Essentially, these are non-functional changes to the service to make it usable by the new project. Another possibility is that the existing service needs enhanced functionality to support the new project. Ideally, the changes can be done in a backward compatible manner; thereby allowing existing consumers of the services to move to the new version deployed to support the new project. Notice that a service versioning strategy should be in place to support these types of changes.

Extensive efforts to modify an existing service may provide less value than building a new, different shared service. Therefore, service modifications are also fed into the service candidate selection process described in the next section. Either type of change to existing services needs to be incorporated into the SOA Roadmap. Obviously, the projects that use the services have a dependency on the service modification activities, and this dependency must be captured in the roadmap.

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BA.070.2 Identify Candidate Services
IP.020.2 Analyze Services

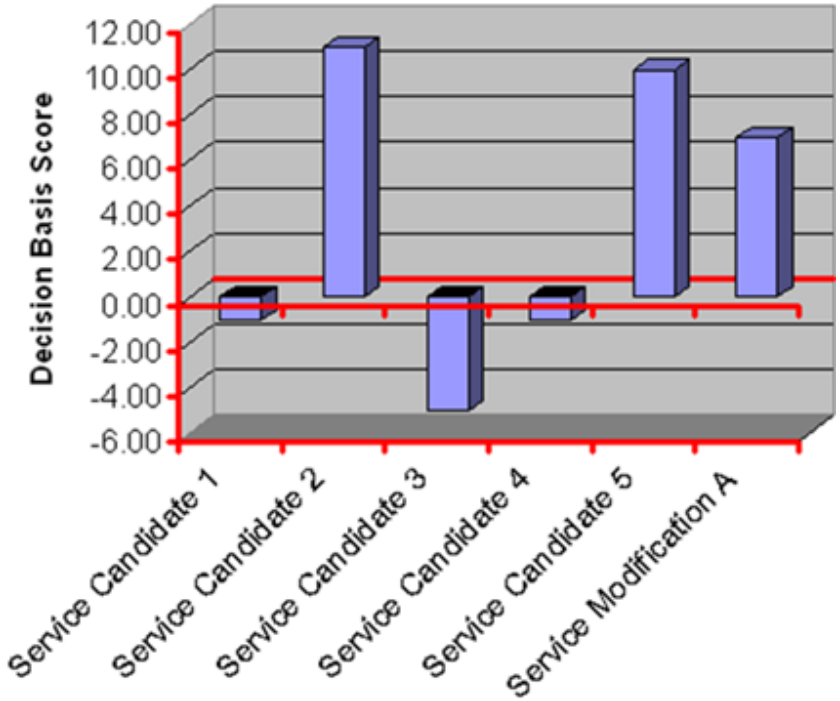
Idetnify Service Candidates

The projects selected for inclusion in the SOA Roadmap can also lead to the creation of new shared services. The projects therefore need to be analyzed to identify the service candidates. There are three primary areas to analyze to identify the service candidates:

1. functional overlap between projects
2. business process tasks that are common across projects
3. common data requirements across projects

Any of these three areas could entail access to an existing system. It is also possible that the overlap between projects is entirely new functionality or data that does not currently exist anywhere in the organization. In this case, the service candidate would be designed to provide this new functionality or data. Service candidate identification and analysis is a large topic in itself, and as such, is beyond the scope of this process. You may find details about this process in the Service Identification and Discovery technique overview. However, once the service candidates have been identified, a selection process should be applied to these similar to the process used for the projects.

Below, you can see an example graphic of service candidate analysis that shows a decision basis score for five service candidates and one service modification:



Service Candidate Analysis

Here a positive **Decision Basis Score** indicates a service candidate that should be realized into a shared service. Thus, service candidates 2, 5, and the Service Modification A should be included in the SOA Roadmap. Service candidates 1, 3, and 4 should not be realized (at this time). The service candidates that are selected are included in the SOA Roadmap. Service candidates (and service modifications) that are not selected should be recorded for future iterations of the roadmap. Although the service candidate did not make the cut for this iteration of the roadmap, it might very well make the cut in a future iteration.

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IP.060 Develop IT Portfolio Plan

Define Schedule

The final step in the SOA Roadmap is to define the schedule for the Roadmap. This schedule should be the schedule for the entire SOA initiative; thus it is the high-level schedule that illustrates the ordering and dependency relationships between program-level activities, the selected projects, service modifications, and the selected service candidates.

The schedule defined here should be the schedule for the entire SOA initiative; thus it is the high-level schedule that illustrates the ordering and dependency relationships between program-level activities, the selected projects, service modifications, and the selected service candidates. The high-level phases of the SOA Roadmap were shown in figure, Increasing Maturity Over Time, that you can see in the supplemental guidance for ER.170 above. However, this figure does not include actual timelines since that could not be determined until the projects, service modifications, and service candidates were identified. The steps to put together the schedule are as follows:

1. Create the program level activities schedule.
2. Create high-level project schedule.
3. Determine delivery dates for service modifications.
4. Determine delivery dates for service candidates.

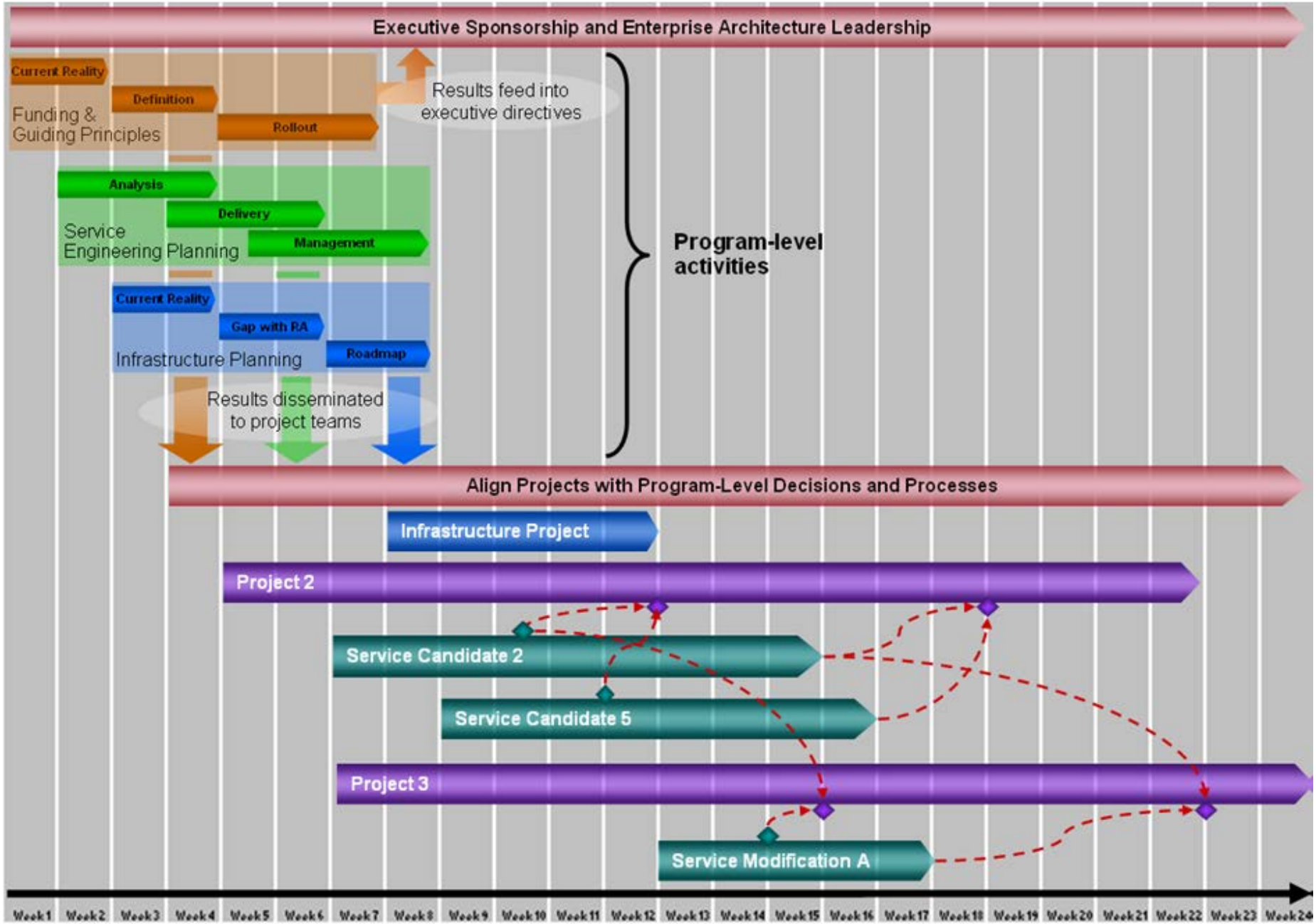
The first activity of defining the schedule is to determine the start date and duration for the program level activities. These are the initial activities put into the SOA Roadmap.

The next step is to determine the start dates for the selected projects. Projects that are expected to leverage the outcomes of the program level activities (usually all projects) should not begin in earnest until the program-level activities complete. Some overlap is allowed since the earliest phases of a project (e.g., Inception) can usually commence before the program-level activities are completed. The end date for the project is determined by effort, complexity, and resource availability. Sometimes the end date is mandated by business needs. In either case, the end date for the project is put into the schedule.

The delivery dates for the service modifications are determined by the needs of the project requiring the changes. Generally, the service modifications must be available in time for project integration testing. Of course, an earlier delivery date is certainly acceptable and is, in fact, preferable.

Finally, the delivery dates for the selected service candidates are determined. These delivery dates are also driven by the needs of the project requiring the new service. Again, the delivery date should meet or beat the date the integration testing for the project will begin. Additionally, the interface definition for the new service(s) should be available to support detailed design for the project.

At this point, any resource constraints need to be incorporated into the schedule. This may require that start dates are delayed or durations may increase. Once these constraints have been included, a schedule can be created and might look something like the schedule shown in the next figure:

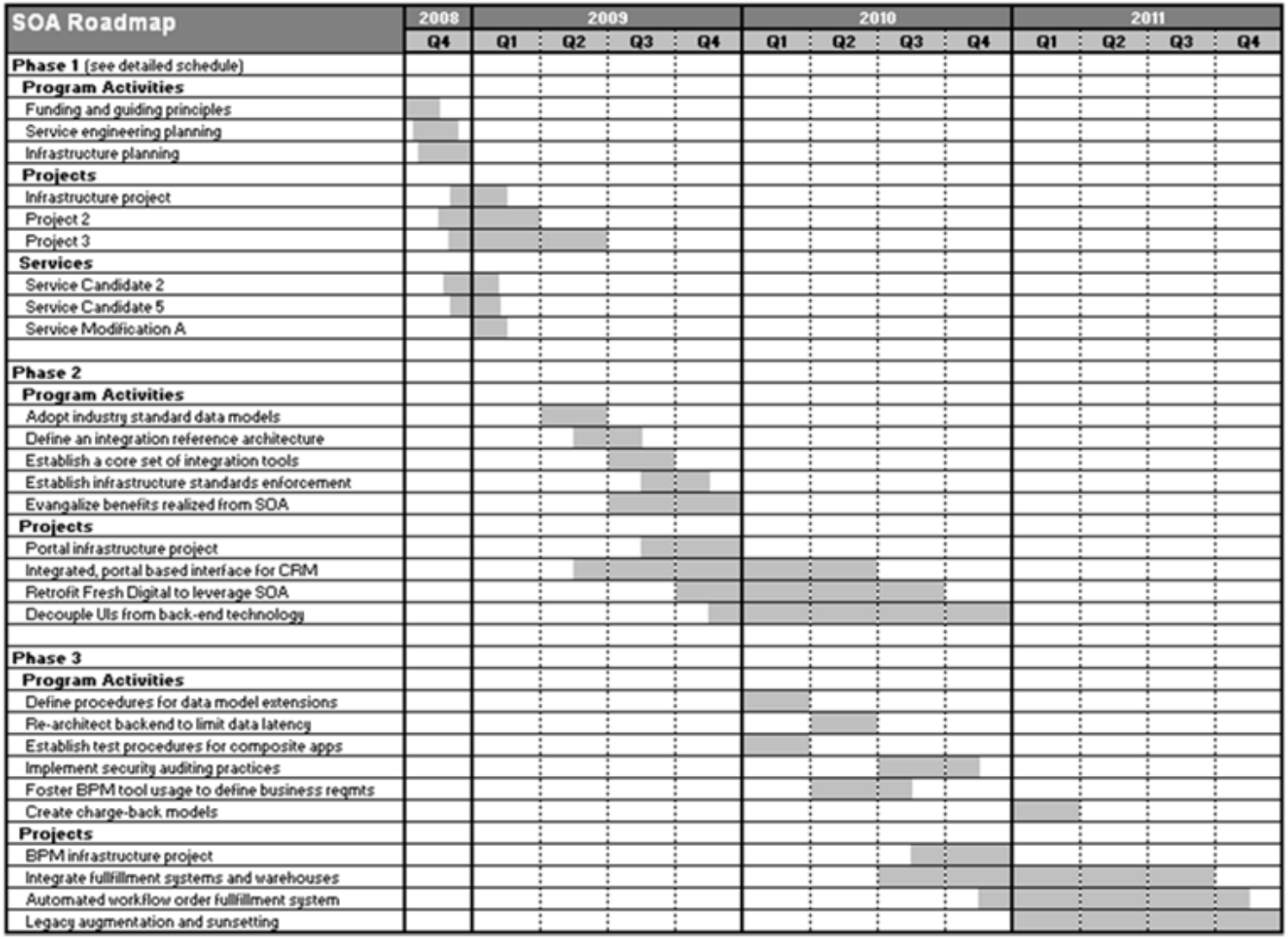


Roadmap Phase 1 Schedule

The above schedule begins with three program-level activities to address the three domains (Business & Strategy, Projects, Portfolios & Services, and Infrastructure) that scored below the Opportunistic maturity level. The results from the program level activities are then disseminated to all the projects and service candidate realization activities shown in the bottom half of the figure. The two projects, two service candidates, and service modification that were selected for inclusion in the roadmap are included in the schedule. The schedule also shows the dependencies between the service realization efforts and the two projects. Both the dependency on the service interface definition and the service deployment are shown since both of these dependencies are important from a scheduling perspective.

The schedule also shows the Infrastructure Project that deploys and configures new components to the service infrastructure needed by the projects and the services. This service infrastructure project is focused purely on realizing the service infrastructure defined in the SOA Reference Architecture. The Infrastructure Planning (that was part of the program level activities) defines what technologies and products are deployed by the Infrastructure Project.

As discussed earlier, the detail in the initial phase of the SOA Roadmap will be much greater than the detail provided for the later phases. The later phases likely include additional program-level activities and more projects and the service candidates on which they are dependent, but will contain less detail. An example of such a schedule is shown in the next figure:



SOA Roadmap Subsequent Phases

The first phase schedule was broken down into weeks but only covered two quarters. This schedule covers the entire planning horizon (three years) but with significantly less detail. The projects chosen for the first phase were relatively short in duration and effort. This is to help ensure early wins for the SOA initiative. Subsequent phases can then afford to undertake more complex efforts (e.g., legacy sun-setting) that could take years to complete.

Subsequent phases will also likely have additional infrastructure projects that continue to build out the reference architecture. The capabilities of the reference architecture

built out in each phase depend on the needs of the projects and services that are included in the phase.

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